

EAST CRAZIES EXAMPLE SOILS SCOPE OF WORK

Date: 2026-05-07

Audience: WLG demonstration package
Purpose: show that the system can produce resource-area scopes of work for a new NEPA project from a structured project intake
Example project: East Crazy Inspiration Divide land exchange
Forest: Custer Gallatin National Forest
District: Bozeman Ranger District
Resource area: Soils

This example assumes a new project is in early planning, before late-stage NEPA materials or resource analysis packages exist. The SOW is generated from the proposed action, project location, preliminary action elements, parcel/trail/access features, likely resource triggers, and standard public source requirements.

For the East Crazies example, the soils scope is driven by:

- proposed new or rerouted trail segments;
- trailhead improvements;
- stream crossings, wetland/riparian approaches, wet soils, hydric soils, erosion, and sediment delivery;
- parcel restrictions, access/easements, grazing-related improvements, and acquired-parcel soil management needs.

The consultant product is a Soils Specialist Report or, if approved by the Forest Service soils reviewer, a short issue-screening memo. Either product must support later EA drafting, Forest Plan consistency documentation, mitigation commitments, and implementation planning.

This is a planning and contracting support SOW. It is not a final agency decision, legal sufficiency review, or NEPA compliance determination.

FIELDWORK

1. Intake And Data Setup

Before fieldwork, compile the new-project intake package:

- proposed action summary, preliminary alternatives if any, project purpose and need, and known action elements;

- conceptual trail, road, access, easement, trailhead, staging, restoration, and parcel boundaries;
- preliminary GIS layers for parcels, routes, crossings, wetlands, riparian areas, streams, slopes, landforms, and administrative boundaries;
- NRCS Web Soil Survey/SSURGO soil map units, erosion hazard, drainage, hydric soil ratings, compaction/rutting limits, and revegetation limitations;
- DEM/slope data, NHD, NWI, MTNHP wetland/riparian data, aerial imagery, grazing allotment/improvement layers, and available landslide or mass-wasting data;
- applicable Forest Plan soil, watershed, road/trail, riparian, recreation, invasive species, and backcountry/recommended-wilderness direction.

Flag missing information early. Typical missing items for a new project are final trail alignments, crossing designs, trailhead footprints, parcel restriction language, construction methods, equipment assumptions, access permission, and GIS source dates.

2. Fieldwork Plan

Submit a fieldwork plan for Forest Service approval before mobilization. Include:

- field questions and decision points;
- activity-area definitions for trail, trailhead, access, wetland/riparian, grazing-improvement, and parcel-management features;
- maps showing proposed review areas, soil map units, slopes, stream/wetland interfaces, route alignments, crossings, parcel boundaries, and access constraints;
- proposed observation points, transects, meander routes, representative soil checks, and photo points;
- method for documenting soil disturbance on linear trail features;
- method for deciding whether FW-STD-SOIL-01 applies, does not apply, or needs a linear-feature explanation;
- GPS/photo naming, field forms, data transfer format, schedule, access needs, safety constraints, and interdisciplinary coordination needs.

3. Field Review Areas

For the East Crazies example, review these areas where access allows:

- proposed Sweet Trunk Trail corridor;
- proposed Inspiration Divide Trail reroute corridor;
- Big Timber Canyon Trailhead improvement area;
- proposed stream crossings, wetland/riparian approaches, seeps/springs, slope breaks, toe slopes, drainage paths, and sediment-routing areas;
- federal disposal parcels and non-federal acquisition parcels where parcel restrictions or future NFS management create soil questions;

- existing roads, trails, route drainage, grazing improvements, compacted areas, eroding areas, disturbed areas, and revegetation/restoration areas.

For another new project, substitute the project-specific action elements and resource trigger areas.

4. Field Methods

Use field methods sufficient to support report-ready conclusions:

- verify representative soil map units, landforms, slope classes, drainage, surface texture, coarse fragments, hydric/wet soil indicators, and erosion susceptibility;
- classify visible soil disturbance, including compaction, rutting, displacement, puddling, surface erosion, organic-matter loss, and mass movement;
- document representative conditions with GPS points, tracks, polygons, and photos;
- use shovel checks, probes, or soil pits where needed for hydric indicators, rooting depth, restrictive layers, wetness, compaction, or restoration feasibility;
- check whether trail and trailhead design can hydrologically disconnect drainage from waterbodies and riparian management zones;
- identify locations needing drainage controls, crossing hardening, wet-weather restrictions, erosion control, revegetation, decompaction, weed prevention, or monitoring.

Detailed lab sampling is not required unless a specific soil question cannot be resolved through soil survey, visual methods, field checks, and professional judgment.

5. Minimum Field Record

For each activity area or representative field check, record:

- date, observer, weather, soil moisture, and access limits;
- activity-area ID, action element, GPS identifier, and photo IDs;
- soil map unit and field-verification notes;
- slope, landform, drainage position, hydric/wet soil indicators, and sediment-connectivity notes;
- existing disturbance type and soil-disturbance class;
- expected disturbance footprint, length, width, acreage, crossing count, or other useful measure;
- BMP, design feature, mitigation, monitoring, data gap, or revisit need.

Fieldwork should occur during snow-free conditions when the soil surface can be observed. Recommend a wet-season or post-storm revisit if runoff, drainage, wetland connectivity, erosion, or sediment delivery cannot be evaluated during the main field visit.

ANALYSIS

1. Existing Condition

Describe the soil setting for affected action areas:

- soil map units, slopes, parent materials, drainage, erosion hazard, compaction/rutting limits, revegetation limits, hydric soil indicators, and sensitive soil attributes;
- existing roads, trails, trailheads, stream crossings, wetland/riparian areas, grazing improvements, disturbed areas, and route drainage;
- existing soil disturbance and sediment connectivity;
- soil conditions on disposal and acquisition parcels where relevant to the land exchange decision;
- data gaps and confidence limits.

2. Proposed-Action Effects

Analyze no action, proposed action, preliminary alternatives if any, and any later Forest Service-provided alternative. Address:

- soil disturbance from trail construction, trail reroute, trailhead work, crossings, access, staging, restoration, and parcel-management changes;
- whether trail work near or across waterbodies could deliver sediment, and what design features would reduce that risk;
- whether a trail reroute would reduce soil or sediment effects by removing an existing stream/wetland crossing;
- whether parcel restrictions, wetland covenants, no-subdivision restrictions, or no-mineral-development restrictions would reduce foreseeable soil disturbance after conveyance;
- whether acquired parcels would create new soil-management needs for NFS management;
- whether trailhead resurfacing, parking, toilet facilities, kiosk construction, or similar improvements would require erosion, compaction, drainage, topsoil, or revegetation controls;
- whether grazing improvements, grazing use, hazardous materials, minerals, vegetation, invasive species, restoration, or carbon assumptions would create soil handling or mitigation needs.

Quantify disturbance where useful. For linear features, use length, width, crossing count, sensitive soil crossings, wetland/riparian approaches, drainage features, and representative disturbance class instead of forcing all effects into an acreage metric.

3. Forest Plan Requirements

Evaluate and document consistency with:

- FW-DC-SOIL-01, FW-DC-SOIL-02, and FW-DC-SOIL-03;
- FW-STD-SOIL-01, including the rationale for applying or screening out the standard;
- FW-GDL-SOIL-01 through FW-GDL-SOIL-08;

- related road/trail direction, especially hydrologic disconnection, mass-wasting avoidance, and hardened stream crossings;
- invasive species, revegetation, riparian, wetland, aquatic, recreation, backcountry, and recommended-wilderness components that affect soil or sediment analysis.

The report must identify which Forest Plan components apply to the proposed action and what design, mitigation, or monitoring requirements should carry forward.

4. Mitigation, BMPs, And Monitoring

Prepare a mitigation/BMP/monitoring matrix with action element, location trigger, soil concern, required measure, responsible party, timing, and verification method. Include, as applicable:

- trail drainage spacing, rolling dips, grade reversals, waterbars, cross drains, armoring, outsloping, and sediment filtering;
- hydrologic disconnection from waterbodies and riparian management zones;
- stream-crossing hardening for trail approaches, streambeds, banks, and approaches;
- avoidance or design modification for wet soils, hydric soils, steep slopes, unstable areas, high mass-wasting potential, and sensitive soil map units;
- wet-weather restrictions, erosion control, decompaction, topsoil/duff salvage, revegetation, mulch, weed-free seed, weed prevention, restoration, and post-construction photo-point monitoring.

5. Cumulative Effects And Uncertainty

Evaluate cumulative soil effects only where there is a useful cause-and-effect relationship. Consider existing and proposed routes, trailheads, unauthorized routes, route drainage, grazing, wetland or riparian disturbance, foreseeable trail or road work, private land development risk after conveyance, and vegetation/restoration actions that could affect ground cover or erosion.

Clearly identify unresolved field limitations, assumptions, and Forest Service decisions needed before soil conclusions can be used in EA drafting, decision support, collection agreements, trail design, or implementation files.

DELIVERABLES

1. Kickoff And Data Request Log

Provide kickoff notes, received and missing data, access constraints, source materials used, stop conditions, and Forest Service decisions needed before fieldwork.

Acceptance criteria:

- identifies missing GIS, trail design, parcel, wetland, route, construction-method, or Forest Plan data;
- identifies whether the work will proceed as a standalone soils report or issue-screening memo;
- identifies field-season and access constraints.

2. Fieldwork Plan

Provide field objectives, activity-area definitions, field maps, observation methods, field forms, data conventions, schedule, safety/access plan, and coordination needs.

Acceptance criteria:

- covers the proposed trail, reroute, trailhead, crossings, wetland/riparian interfaces, parcel restrictions, and other project-specific soil trigger areas;
- explains how soil disturbance will be documented for linear features;
- identifies whether wet-season or post-storm verification may be needed.

3. Field Data Package

Provide GPS data, completed field forms, photo log, soil-check notes, soil map unit verification, wetland/riparian and sediment-connectivity notes, and a short field findings memo.

Acceptance criteria:

- every material field conclusion is traceable to location, date, observer, photo, and activity area;
- sensitive soil, hydric/wet soil, erosion, compaction, rutting, displacement, drainage, and sediment concerns are mapped or explicitly screened out;
- data are suitable for Custer Gallatin GIS and ID-team use.

4. Analysis Tables And Map Atlas

Provide:

- soil map unit and activity-area crosswalk;
- existing and expected disturbance table;
- action-element by soil concern matrix;
- stream/wetland crossing and sediment-connectivity table;
- mitigation/BMP/monitoring matrix;
- Forest Plan requirements matrix;
- data-gap and uncertainty table;
- project, soil-analysis area, soil map unit, slope/erosion hazard, wetland/riparian, road/trail, observation, and mitigation maps.

Acceptance criteria:

- maps include scale, north arrow, coordinate system, data date, project boundary, activity-area IDs, and source attribution;
- tables distinguish no action, proposed action, preliminary alternatives, and any later analyzed alternative;
- linear-feature effects use useful measures and do not rely on misleading acreage-only metrics.

5. Draft Soils Specialist Report Or Issue-Screening Memo

Provide a draft report or approved issue-screening memo with affected environment, methods, spatial and temporal scale, resource indicators, direct and indirect effects, cumulative effects, mitigation/BMPs/monitoring, Forest Plan requirements, references, and appendices.

Acceptance criteria:

- directly addresses the proposed action and each soil-triggering action element;
- states whether FW-STD-SOIL-01 applies or is screened out for trail work;
- ties conclusions to field evidence, GIS evidence, public source requirements, or specialist judgment.

6. EA-Ready Soils Text

Provide concise text for the future EA or appendix covering affected environment, environmental consequences, mitigation/design criteria, Forest Plan requirements, and cumulative effects if needed.

Acceptance criteria:

- suitable for public disclosure;
- references the soils report or issue-screening memo instead of duplicating all technical detail;
- identifies design criteria or monitoring that must carry into later decision or implementation files.

7. Final Report And Data Package

Provide final report or memo, final PDF, editable source file, comment-response log, final tables, map atlas, GIS layers or geodatabase/geopackage, photo log, field forms, source data inventory, metadata/readme, and data limitations memo.

Acceptance criteria:

- resolves Forest Service and ID-team comments;
- preserves source citations and field-data traceability;
- documents residual uncertainty and remaining Forest Service decisions;
- final PDF starts with a valid ‘%PDF-’ header.

SOURCES CITED

- 36 CFR 220.4, Forest Service NEPA general requirements:
<https://ecfr.io/Title-36/Section-220.4>
- 36 CFR 220.7, Environmental assessment and decision notice:
<https://ecfr.io/Title-36/Section-220.7>
- USDA Forest Service Manual, Region 1 Supplement, FSM 2550 Soil Management.
- Custer Gallatin National Forest Land Management Plan:
<https://www.fs.usda.gov/r01/custergallatin/planning/forest-plan/custer-gallatin-land-management-plan-forest-plan-revision>
- Forest Service Soil Quality Monitoring resources:
<https://www.fs.usda.gov/soils/monitoring.shtml>
- Forest Service Soil-Disturbance Field Guide:
https://www.fs.usda.gov/t-d/php/library_card.php?p_num=0819+1815P
- NRCS Web Soil Survey and SSURGO:
<https://www.nrcs.usda.gov/resources/data-and-reports/web-soil-survey>
- Forest Service National Best Management Practices Program:
<https://www.fs.usda.gov/naturalresources/watershed/bmp.shtml>
- East Crazies example project intake assumptions: land exchange, proposed trail construction, proposed trail reroute, trailhead improvements, parcel restrictions, easements, wetland/riparian interfaces, grazing-related improvements, and acquired-parcel management needs.